ChemPubSoc Europe

DOI: 10.1002/chem.201503278

CHEMISTRY
A European Journal
Communication

Catalysis

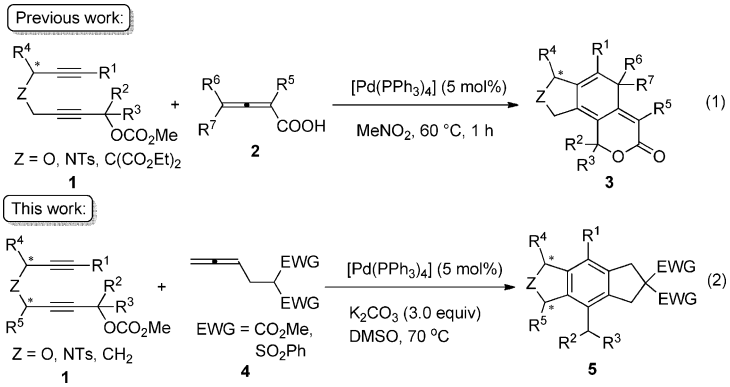
Palladium-Catalyzed Triple Cyclization of 2,7-Alkadiynylic Carbonates with Allenes Bearing a Carbon Nucleophile

Xin Huang,[a] Wangteng Wu,[a] Chunling Fu,[a] Yihua Yu,[b] and Shengming Ma*[a]

Dedicated to Dr. Negishi on the occasion of his 80th birthday

Abstract: Palladium-catalyzed tandem reactions of 2,7-alkadiynylic carbonates with allenes bearing a carbon nucleophile such as malonate and bis(phenylsulfonyl)methane for the efficient syntheses of fused tricycles, such as 3,5,6,7-tetrahydro-*1H*-indeno[5,6-*c*]furan, 1,2,3,5,6,7-hexahydrocyclopenta[*f*]isoindole, and 1,2,3,5,6,7-hexahydro-*s*-indacene derivatives, have been reported. The reaction forms two five-membered rings and one benzene ring fusing them in the middle.

Polycycles are important structural motifs existing in nature with significant pharmacological and biological activities[1] and thus have attracted much attention from the chemical community.[2] However, efficiency is always the goal. In our previous work, an efficient approach to substituted 1,5,7,8,9-pentahydrocyclopenta-[*h*]-2-benzopyran-3-one derivatives by a palladium-catalyzed tandem reaction of 2,7-alkadiynylic carbonates[3] with 2,3-allenoic acids has been reported (Scheme 1, Eq. (1)).[4] We envisioned that similar tandem reactions may occur if the 2,3-allenoic acids are replaced with allenes bearing a carbon-centered nucleophile. However, a totally different tricyclic skeleton, that is, 3,5,6,7-tetrahydro-*1H*-indeno[5,6-*c*]furan, 1,2,3,5,6,7-hexahydrocyclopenta[*f*]isoindole, or 1,2,3,5,6,7-hexahydro-*s*-indacene derivatives, was constructed (Scheme 1, Eq. (2)). Herein, we wish to report such recent observations.



Scheme 1. Different tandem reactions of 2,7-alkadiynylic carbonates with 2,3-allenoic acids or allene bearing a carbon-centered nucleophile.

Initially, we conducted the reaction of 2,7-alkadiynylic carbonate (**1a**) (0.3 mmol) and dimethyl 2-(buta-2,3-dien-1-yl)malonate (**4a**) (1.2 equiv) under the catalysis of $[Pd(PPh_3)_4]$ (5 mol%) and K_3PO_4 (3.0 equiv) at 70 °C for 12 h in DMSO. Unexpectedly, we did not observe the predicted tricyclic product **3aa**, instead, a new tricyclic 4-isopropyl-6,6-dimethoxycarbonyl-8-phenyl-3,5,6,7-tetrahydro-1*H*-indeno[5,6-*c*]furan (**5aa**) was identified in 67% NMR yield and 53% isolated yield as determined by the ^1H and ^{13}C NMR spectroscopic analysis (Table 1, entry 3). Encouraged by this exciting observation, the influence of the critical reaction parameters was investigated and representative results are summarized in Table 1.

By changing the molar ratio of **1a**/**4a**, all the reactions afforded **5aa** in lower yields of 59–65% (Table 1, entries 1–2 and 4,). After screening a series of solvents, it was found that DMSO was still the best one. We could not observe any cyclic product **5aa** when using 1,4-dioxane or THF as the solvent (entries 5 and 6). In contrast, the reactions in CH_3CN or DMF led to poorer results than that in DMSO (entries 7 and 8). The effect of base was also investigated with K_2CO_3 being slightly better than K_3PO_4 and **5aa** could be formed in the highest yield of 69% (entries 9 and 10). In addition, running the reaction at 60 or 80 °C failed to improve the yield of **5aa** (entries 11 and 12). Therefore, $[Pd(PPh_3)_4]$ (5 mol%), **4a** (1.2 equiv), and K_2CO_3 (3.0 equiv) in DMSO at 70 °C were established as the standard reaction conditions for further study.

With the optimized conditions in hand, we first examined the reactivity of various oxygen-tethered 2,7-alkadiynylic carbonates **1** with dimethyl 2-(buta-2,3-dien-1-yl)malonate (**4a**) or 1,1-bis(phenylsulfonyl)penta-3,4-diene (**4b**) affording different 3,5,6,7-tetrahydro-*1H*-indeno[5,6-*c*]furan derivatives (Table 2, entries 1–16). The reaction of tertiary 2,7-diynic carbonates

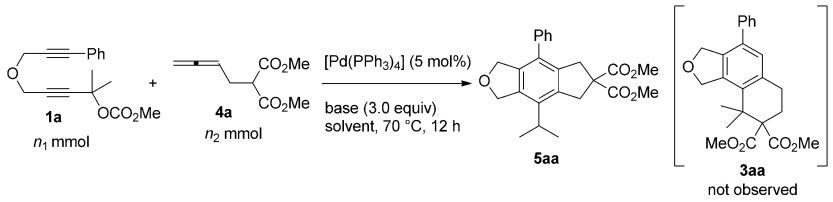
[a] Dr. X. Huang, W. Wu, Prof. Dr. C. Fu, Prof. Dr. S. Ma
Laboratory of Molecular Recognition and Synthesis
Department of Chemistry, Zhejiang University
Hangzhou 310027, Zhejiang (P. R. China)
E-mail: masm@sioc.ac.cn

[b] Prof. Dr. Y. Yu
Shanghai Key Laboratory of Magnetic Resonance
Department of Physics, East China Normal University
Shanghai 200062 (P. R. China)

Supporting information for this article is available on the WWW under http://dx.doi.org/10.1002/chem.201503278.

© 2015 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

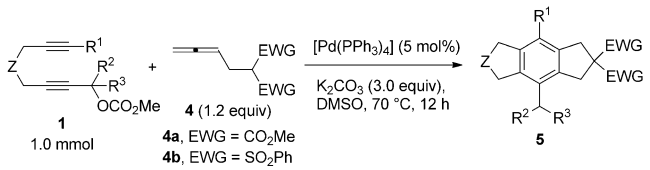
Table 1. Optimization of reaction conditions.[a]



Entry	n_1/n_2	Base	Solvent	NMR yield of **5aa** [%][b]
1	1.2:1	K_3PO_4	DMSO	63
2	1:1	K_3PO_4	DMSO	65
3	1:1.2	K_3PO_4	DMSO	67 (53)[c]
4	1:1.5	K_3PO_4	DMSO	59
5	1:1.2	K_3PO_4	Dioxane	0
6	1:1.2	K_3PO_4	THF	0
7	1:1.2	K_3PO_4	CH_3CN	19
8	1:1.2	K_3PO_4	DMF	50
9	1:1.2	Cs_2CO_3	DMSO	27
10	**1:1.2**	**K_2CO_3**	**DMSO**	**69**
11[d]	1:1.2	K_2CO_3	DMSO	64
12[e]	1:1.2	K_2CO_3	DMSO	57

[a] Reaction conditons: **1a** (0.3 mmol), $[Pd(PPh_3)_4]$ (5 mol%), base (3.0 equiv) at 70 °C in solvent (3.0 mL) for 12 h unless otherwise noted. [b] Determined by ^1H NMR spectroscopic analysis of the crude products with mesitylene as the internal standard. [c] Isolated yield. [d] The reaction was conducted at 60 °C for 22 h. [e] The reaction was conducted at 80 °C for 4 h.

Table 2. Pd-catalyzed tandem reactions of 2,7-alkadiynylic carbonates **1** with 2,3-allenyl carbon nucleophile **4**.[a]



Entry	1				4	Yield of 5 [%][b]
	Z	R^1	R^2	R^3		
1	O	Ph	Me	Me (**1a**)	**4a**	63 (**5aa**)
2[c]	O	Ph	Me	Me (**1a**)	**4a**	72 (**5aa**)
3	O	Ph	Me	Me (**1a**)	**4b**	75 (**5ab**)
4	O	Ph	Et	Et (**1b**)	**4a**	67 (**5ba**)
5	O	Ph	$-(CH_2)_4-$ (**1c**)		**4a**	48 (**5ca**)
6	O	Ph	$-(CH_2)_{11}-$ (**1d**)		**4a**	65 (**5da**)
7	O	Ph	Me	Pr (**1e**)	**4a**	64 (**5ea**)
8	O	Ph	Pr	H (**1f**)	**4b**	60 (**5fb**)
9	O	Ph	Ph	H (**1g**)	**4b**	46 (**5gb**)
10	O	*p*-ClC_6H_4	Me	Me (**1h**)	**4a**	49 (**5ha**)
11	O	*p*-$MeOC_6H_4$	Me	Me (**1i**)	**4a**	50 (**5ia**)
12	O	*p*-MeC_6H_4	Me	Me (**1j**)	**4a**	59 (**5ja**)
13	O	*o*-MeC_6H_4	$-(CH_2)_4-$ (**1k**)		**4a**	50 (**5ka**)
14	O	*m*-MeC_6H_4	Me	Et (**1l**)	**4a**	68 (**5la**)
15	O	2-thienyl	Pr	H (**1m**)	**4b**	60 (**5mb**)
16	O	cyclohexenyl	Me	Me (**1n**)	**4a**	45 (**5na**)
17	NTs	Ph	Me	Me (**1o**)	**4a**	43 (**5oa**)
18[d]	NTs	Ph	$-(CH_2)_5-$ (**1p**)		**4a**	53 (**5pa**)
19	CH_2	Ph	Me	Me (**1q**)	**4a**	67 (**5qa**)

[a] Reaction conditions: **1** (1.0 mmol), **4a** (1.2 equiv), $[Pd(PPh_3)_4]$ (5 mol%), K_2CO_3 (3.0 equiv) at 70 °C in DMSO (10.0 mL) for 12 h unless otherwise noted. [b] Isolated yield. [c] The reaction was conducted on a 1 g scale of **1a**. [d] The reaction time was 24 h.

(**1a**–**e**) proceeded smoothly affording the corresponding tricycles **5aa**–**ea** in moderate yields of 48–75%, no matter if the R^2 and R^3 groups were the same or not (entries 1–7). Also, the reaction of secondary 2,7-alkadiynylic carbonates (**1f**, **1g**) with **4b** afforded the tricycles **5fb** and **5gb** in 60 and 46% yields, respectively (entries 8 and 9). The carbon–carbon triple bond at the 8-position of the 2,7-alkadiynylic carbonates **1** may be substituted with an aryl group bearing either electron-deficient or -rich groups (entries 10–14). In addition, this method could be extended to 2-thienyl- or cyclohexenyl-substituted 2,7-alkadiynylic carbonates (entries 15 and 16). It is also applicable to nitrogen- or carbon-tethered 2,7-alkadiynylic carbonates to form 1,2,3,5,6,7-hexahydrocyclopenta[*f*]isoindole derivatives (**5oa**, **5pa**) and 1,2,3,5,6,7-hexahydro-*s*-indacene derivative **5qa** in moderate yields (entries 17–19). We also investigated the reaction of the reaction of **1a** with **4a** on a 1 g scale (entry 2).

The structure of this tricyclic product was unambiguously established by the X-ray diffraction analysis of product **5ia** (Figure 1).[5]

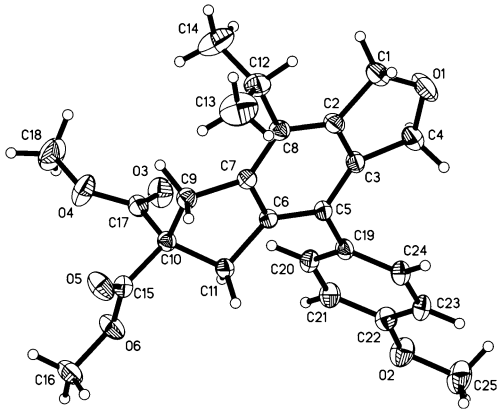


Figure 1. ORTEP representation of product **5ia**.

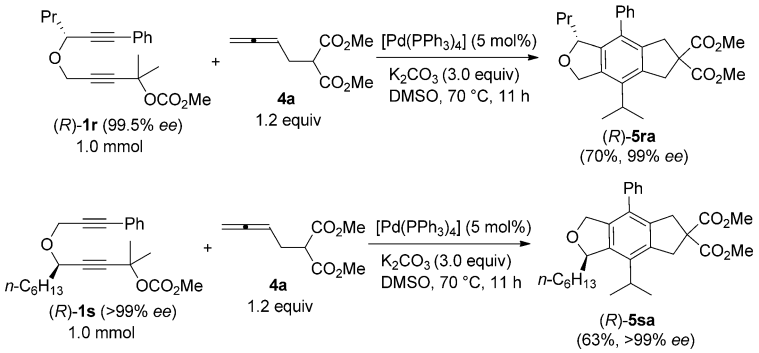
When the optically active 2,7-alkadiynylic carbonates (*R*)-**1r** and (*R*)-**1s** containing a central chirality were subjected to the standard reaction conditions, (*R*)-**5ra** and (*R*)-**5sa** were obtained in 70 and 63% yields, respectively, without any loss of enantiomeric purity (Scheme 2).

Interestingly, the substrate **1t**, which has two symmetric yne-propargylic carbonate units, could also react with **4a** affording the corresponding product **5ta** in 46% yield. Six rings were constructed in just one operation [Eq. (3)].

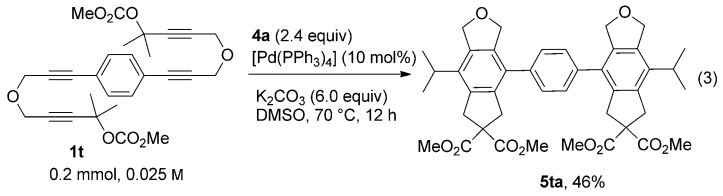
On the basis of these results, we propose a possible mechanism as shown in Scheme 3 by taking the reaction of **1a** with

© 2015 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

15213765, 2015, 44, Downloaded from https://chemistry-europe.onlinelibrary.wiley.com/doi/10.1002/chem.201503278 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

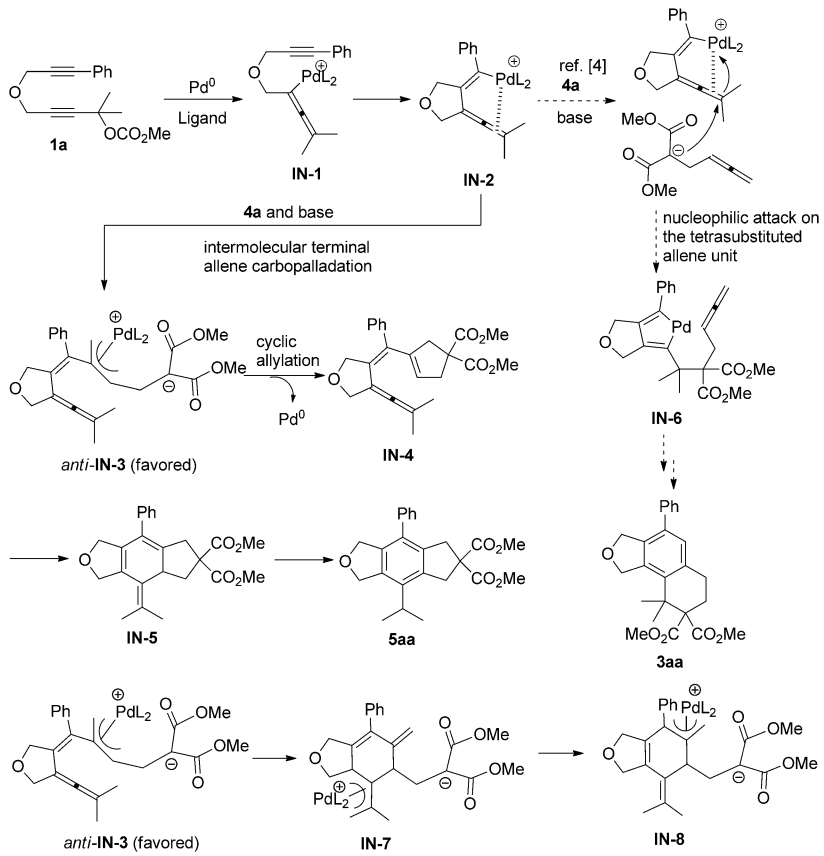


Scheme 2. Synthesis of optically active tricyclic products (*R*)-**5ra** and (*R*)-**5sa**.



4a as an example: oxidative addition of **1a** with Pd^0 would afford the allenylpalladium intermediate **IN-1**,[6] which undergoes intramolecular insertion of the triple bond to generate the alkenylpalladium species **IN-2**.[3] The species **IN-2** undergoes highly regioselective intermolecular carbopalladation of the allene moiety in **4a** to generate allylpalladium intermediate *anti*-**IN-3** highly stereoselectively,[7] instead of forming the key intermediate **IN-6** by the direct attack of the carbon nucleophile leading the formation of **3aa** as reported in ref. [4] probably due to the steric effect and/or the weaker nucleophilicity of the in situ generated anionic carbon center. Then intramolecular nucleophilic allylation forms allene–diene intermediate **IN-4** and regenerates the catalytically active Pd^0. After 6*e*-electrocyclization and aromatization,[8] the final product **5aa** was delivered. There is another alternative pathway to **IN-5**. Intramolecular carbopalladation of allene would form a new allylpalladium intermediate **IN-7**. Due to the presence of the conjugated diene, **IN-7** would be transformed to **IN-8** with the allylpalladium unit attacked by the nucleophile to form the final five-membered ring in **IN-5**.

In conclusion, we have developed an efficient methodology for the syntheses of fused tricycles, such as 3,5,6,7-tetrahydro-1*H*-indeno[5,6-*c*]furan, 1,2,3,5,6,7-hexahydrocyclopenta[*f*]isoindole, and 1,2,3,5,6,7-hexahydro-*s*-indacene derivatives, from 2,7-alkadiynylic carbonates with allenes bearing a carbon nucleophile. Four new carbon–carbon bonds are formed within one operation from readily available starting materials.[9] No racemization occurred to the central chirality in the starting materials. Due to the importance of this attractive tricyclic skeleton,[10] the reaction will be potentially useful in organic synthesis and medical chemistry. Further studies on expanding the scope of the reaction and synthetic applications are being pursued in our laboratory.



Scheme 3. Proposed mechanism for the formation of **5aa**.

Experimental Section

General

To a flame-dried Schlenk tube containing K_2CO_3 (413.9 mg,

© 2015 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

15213765, 2015, 44, Downloaded from https://chemistry-europe.onlinelibrary.wiley.com/doi/10.1002/chem.201503278 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

15213765, 2015, 44, Downloaded from https://chemistry-europe.onlinelibrary.wiley.com/doi/10.1002/chem.201503278 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

3.0 mmol) were added $[Pd(PPh_3)_4]$ (57.7 mg, 0.05 mmol), **1a** (285.5 mg, 1.0 mmol)/DMSO (5.0 mL), and **4a** (220.9 mg, 1.2 mmol)/DMSO (5.0 mL) sequentially under a nitrogen atmosphere. The reaction was complete after being stirred at 70 °C for 12 h as monitored by TLC analysis (eluent: petroleum ether/ethyl acetate = 10:1). The resulting mixture was quenched with an aqueous solution of diluted hydrochloric acid ((*v*/*v* = 10%, commercially available concentrated hydrochloric acid was diluted with water according the volume ratio), 10 mL) slowly and extracted with ethyl acetate (30 mL × 3). The combined organic layer was washed with water and brine, and dried over anhydrous Na_2SO_4. After filtration and evaporation of the solvent, chromatography on silica gel (eluent: petroleum ether/ethyl acetate = 15:1) afforded **5aa** (249.0 mg, 63%) as a white solid. M.p. 144–145 °C (*n*-hexane/CH_2Cl_2); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.46–7.20 (m, 5H, ArH), 5.19 (s, 2H, CH_2), 4.93 (s, 2H, CH_2), 3.71 (s, 6H, Me × 2), 3.67 (s, 2H, CH_2), 3.47 (s, 2H, CH_2), 3.14–2.93 ppm (m, 1H, CH), 1.28 (d, J = 7.2 Hz, 6H, Me × 2); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 171.8, 138.3, 138.1, 137.6, 137.4, 136.5, 136.1, 129.7, 128.4, 128.1, 127.2, 73.0, 72.6, 60.3, 52.8, 39.4, 39.1, 31.2, 21.1 ppm; IR (KBr): $\tilde{\nu}$ = 3054, 2955, 2901, 2856, 1732, 1601, 1459, 1432, 1365, 1275, 1242, 1203, 1162, 1066, 1040 cm^{-1}; MS (EI): *m*/*z* (%) 394 (100) $[M]^+$; calcd (%) for $C_{24}H_{26}O_5$: C 73.08, H 6.64; found: C 72.80, H 6.61.

Acknowledgements

Financial support from the National Natural Science Foundation of China (21232006) and National Natural Basic Research Program of China (2015CB856600) are greatly appreciated. We thank Mr. H. Luo in this group for reproducing the preparation of **5mb**, and (*R*)-**5ra**. S. Ma is a Qiu Shi Adjunct Professor at Zhejiang University.

Keywords: alkadiynylic carbonates · allenes · carbon nucleophiles · palladium · triple cyclization

[1] For selected reviews, see: a) M. Inoue, *Chem. Rev.* **2005**, *105*, 4379; b) M. Randić, *Chem. Rev.* **2003**, *103*, 3449; c) T. Nakata, *Chem. Rev.* **2005**, *105*, 4314; d) M. Malacria, *Chem. Rev.* **1996**, *96*, 289.

[2] For selected recent reviews on tandem reactions to construct polycycles, see: a) J.-C. Wasilke, S. J. Obrey, R. T. Baker, G. C. Bazan, *Chem. Rev.* **2005**, *105*, 1001; b) K. C. Nicolaou, D. J. Edmonds, P. G. Bulger, *Angew. Chem. Int. Ed.* **2006**, *45*, 7134; *Angew. Chem.* **2006**, *118*, 7292; c) H.-C. Guo, J.-A. Ma, *Angew. Chem. Int. Ed.* **2006**, *45*, 354; *Angew. Chem.* **2006**, *118*, 362; d) A. Padwa, S. K. Burb, *Tetrahedron* **2007**, *63*, 5341; e) T. Vlaar, E. Ruijter, R. V. A. Orru, *Adv. Synth. Catal.* **2011**, *353*, 809; f) L. Lu, J. Chen, W. Xiao, *Acc. Chem. Res.* **2012**, *45*, 1278; g) T. Lechel, F. Pfrengle, H.-U. Ressig, R. Zimmer, *ChemCatChem.* **2013**, *5*, 2100;h) B. Alcaide, P. Almendros, C. Aragoncillo, *Chem. Soc. Rev.* **2014**, *43*, 3106.

[3] For selected reports on palladium-catalyzed cyclization reactions of ene- or yne-propargylic carbonates, see: a) R. Grigg, R. Rasul, J. Redpath, D. Wilson, *Tetrahedron Lett.* **1996**, *37*, 4609; b) W. Oppolzer, A. Pimm, B. Stammen, W. E. Hume, *Helv. Chim. Acta* **1997**, *80*, 623; c) R. Grigg, R. Rasul, V. Savic, *Tetrahedron Lett.* **1997**, *38*, 1825; d) A. G. Steinig, A. de Meijere, *Eur. J. Org. Chem.* **1999**, 1333; e) J. Böhmer, R. Grigg, J. D. Marchbank, *Chem. Commun.* **2002**, 768; f) F. Wang, X. Tong, J. Cheng, Z. Zhang, *Chem. Eur. J.* **2004**, *10*, 5338; g) S. Zhao, K. Ji, L. Lu, T. He, A. Zhou, R. Yan, S. Ali, X. Liu, Y. Liang, *J. Org. Chem.* **2012**, *77*, 2763.

[4] X. Lian, S. Ma, *Angew. Chem. Int. Ed.* **2008**, *47*, 8255; *Angew. Chem.* **2008**, *120*, 8379.

[5] Crystal data for compound **5ia**: $C_{25}H_{28}O_6$; *MW* = 424.47; monoclinic, space group *C2/c*; final *R* indices [$I > 2\sigma(I)$], *R*1 = 0.0501, *wR*2 = 0.1282, *R* indices (all data) *R*1 = 0.0755, *wR*2 = 0.1469; *a* = 24.463(3), *b* = 13.1036(7), *c* = 15.3564(18) Å; α = 90.0, β = 116.330(14), γ = 90.0°; *V* = 4411.9(8) Å^3; *T* = 293(2) K; *Z* = 8; reflections collected/unique 9670/4035 (R_{int} = 0.0296); number of observations [$> 2\sigma(I)$] 2862, parameters: 285. CCDC-1415089 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.

[6] For selected reviews, see: a) J. Tsuji, T. Mandai, *Angew. Chem. Int. Ed. Engl.* **1995**, *34*, 2589; *Angew. Chem.* **1995**, *107*, 2830; b) L. Guo, X. Duan, Y. Liang, *Acc. Chem. Res.* **2011**, *44*, 111; c) J. Ye, S. Ma, *Acc. Chem. Res.* **2014**, *47*, 989.

[7] a) S. Ma, *Acc. Chem. Res.* **2003**, *36*, 701; b) S. Ma, N. Jiao, S. Zhao, H. Hou, *J. Org. Chem.* **2002**, *67*, 2837.

[8] a) W. Reischl, W. H. Okamura, *J. Am. Chem. Soc.* **1982**, *104*, 6115; b) P. Turnbull, H. Moore, *J. Org. Chem.* **1995**, *60*, 3274; c) T. Choshi, T. Sada, H. Fujimoto, C. Nagayama, E. Sugino, S. Hibino, *Tetrahedron Lett.* **1996**, *37*, 2593; d) T. M. Morwick, L. A. Paquette, *J. Org. Chem.* **1997**, *62*, 627; e) T. Choshi, T. Sada, H. Fujimoto, C. Nagayama, E. Sugino, S. Hibino, *J. Org. Chem.* **1997**, *62*, 2535; f) H. Hagiwara, T. Choshi, H. Fujimoto, E. Sugino, S. Hibino, *Chem. Pharm. Bull.* **1998**, *46*, 1948; g) H. Hagiwara, T. Choshi, H. Fujimoto, E. Sugino, S. Hibino, *Tetrahedron* **2000**, *56*, 5807; h) H. Hagiwara, T. Choshi, H. Fujimoto, S. Hibino, *Chem. Pharm. Bull.* **2001**, *49*, 881; i) G. Zhao, Q. Zhang, H. Zhou, *Adv. Synth. Catal.* **2013**, *355*, 3492.

[9] For details of the syntheses of starting materials see the Supporting Information.

[10] For selected reports on natural products containing this tricyclic skeleton, see: a) A. Arnone, G. Candiani, G. Nasini, R. Sinisi, *Tetrahedron* **2003**, *59*, 5033; b) Y. Aoyagi, A. Yamazaki, R. Kato, F. Tobe, H. Fukaya, T. Nishikawa, A. Nakahashi, N. Miura, K. Monde, K. Takeya, *Tetrahedron Lett.* **2011**, *52*, 1851; c) C. L. Hugelshofer, T. Magauer, *Angew. Chem. Int. Ed.* **2014**, *53*, 11351; *Angew. Chem.* **2014**, *126*, 11533.

Received: August 18, 2015
Published online on September 11, 2015

© 2015 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim